PHASE 7

USER EVALUATION AND RESULTS

Static Explanation Baseline versus Interactive Power BI Dashboard

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

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Document status
This document presents the completed Phase 7 evaluation. The numerical findings are calculated from the participant response dataset used in the study.

Contents

Executive Summary

Phase 7 evaluates whether the interactive Power BI report developed in Phase 6 offers a practical advantage over a static presentation of the same churn predictions and SHAP explanations. A within-subject comparison was designed so that each participant could use both conditions and complete equivalent tasks.

The current dataset contains 10 participant profiles and twenty condition-level records. The interactive dashboard produced a higher mean task accuracy (91.67% compared with 70.83%) and a higher mean overall rating (4.45 compared with 3.53 out of 5). Eight participants preferred the interactive dashboard, one preferred the static baseline, and one reported no clear preference.

The strongest reported benefits of interactivity were easier factor identification, improved comparison of customers and segments, and stronger support for retention decisions. The static version remained useful for a quick overview and required no navigation experience.

1. Phase Overview

The earlier phases of the thesis produced a tuned Gradient Boosting churn model, probability-based SHAP explanations, structured customer-level outputs, and a six-page Power BI dashboard. Phase 7 examines how these outputs are understood and used when they are presented as either fixed pages or an interactive visual analytics system.

The evaluation was intentionally kept compact because its purpose is to demonstrate the assessment of the completed system without creating an unnecessarily large set of study documents. The Phase 7 materials therefore consist of one evaluation framework, one static baseline and task document, one combined form, participant response sheets, and one analysis report.

2. Evaluation Objective and Systems

2.1 Evaluation objective

The main objective is to determine whether interactivity improves the interpretation and practical use of churn predictions and explanations. The study considers task accuracy, completion time, understanding, trust, decision support, ease of use, usefulness, and final system preference.

2.2 Condition A - Static explanation baseline

The static condition presents fixed dashboard screenshots. Users can read the model metrics, risk information, SHAP explanations, segment comparisons, and threshold results, but they cannot change filters or select another customer.

2.3 Condition B - Interactive Power BI dashboard

The interactive condition uses the complete Power BI report. Users can move between pages, use slicers, select customers, compare segments, and inspect explanations that update according to their selections.

Fair comparison principle
Both conditions use the same underlying customer records, prediction outputs, SHAP values, and model-performance results. The primary experimental difference is the availability of interaction.

3. Research Questions and Hypotheses

- RQ1: Does the interactive dashboard improve understanding of churn predictions compared with the static baseline?
- RQ2: Does interactivity help users identify important global and customer-specific churn factors?
- RQ3: Does the dashboard provide stronger support for customer-retention decisions?
- RQ4: Does the dashboard increase confidence and trust in the model output?
- RQ5: Is the dashboard perceived as more usable and useful than the static baseline?

Hypothesis	Expected outcome
H1	Higher task accuracy in the interactive condition.
H2	Higher understanding ratings for the interactive dashboard.
H3	Higher trust and confidence ratings for the interactive dashboard.
H4	Higher ease-of-use and usefulness ratings for the interactive dashboard.
H5	Better support for retention decisions in the interactive condition.
H6	Interactivity may require additional exploration time but should support more complete answers.

4. Study Design and Participants

4.1 Within-subject design

Each participant uses both systems. Equivalent tasks are completed under each condition, which allows direct comparison within the same participant group. To reduce order effects, P01-P05 use the static version first, while P06-P10 use the interactive dashboard first.

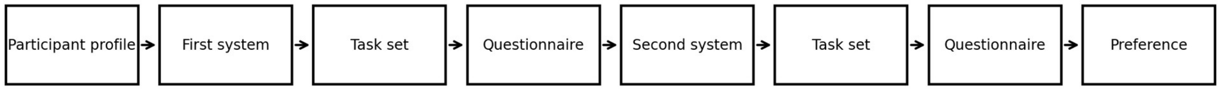

Figure 1. Compact evaluation workflow used in Phase 7.

4.2 Participant profiles

ID	Name	Age	Background	Power BI experience	Order
P01	Lenin	23	Master's in Data Science	Basic	Static first
P02	Aahan	23	Pilot	None	Static first
P03	Steve	24	Master's in Cybersecurity	None	Static first
P04	Smith	24	Master's in Cybersecurity	None	Static first
P05	Nathen	24	Bachelor's in Data Science	Basic	Static first
P06	Deva	24	Master's in Data Science	Intermediate	Interactive first
P07	Deekshi	27	Master's in Architecture	None	Interactive first
P08	Helaine	23	Master's in Physics	None	Interactive first
P09	Akshata	23	Master's in Data Science	Intermediate	Interactive first
P10	Kshitija	23	Master's in Data Science	Intermediate	Interactive first

The group covers data science, cybersecurity, architecture, physics, aviation, and professional pilot training. The mixture supports an initial view of how the report may be interpreted by users with different technical backgrounds and different levels of Power BI familiarity.

5. Static Baseline and Evaluation Tasks

The static baseline was prepared from fixed screenshots of the six Power BI pages. This preserves the information content of the dashboard while removing filtering, selection, and drill-down behaviour.

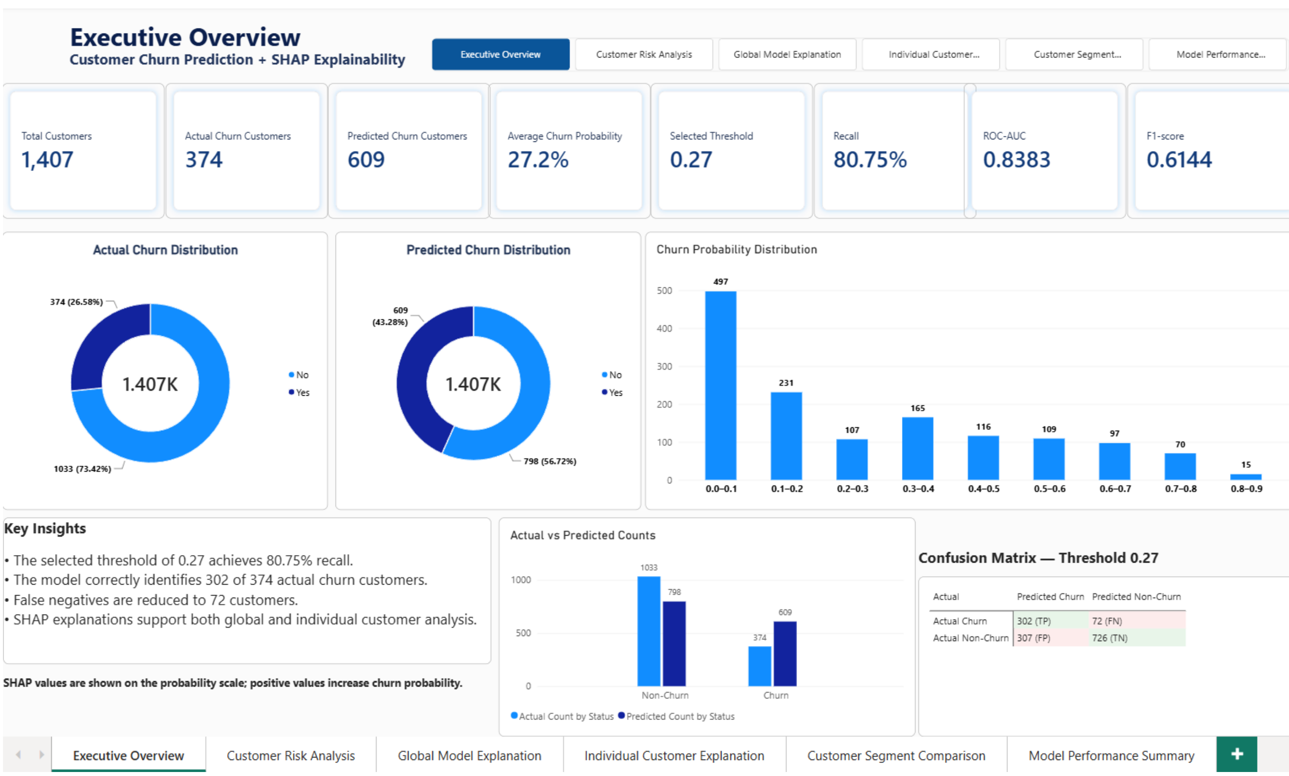

Figure 2. Static Executive Overview used to answer overall churn and prediction questions.

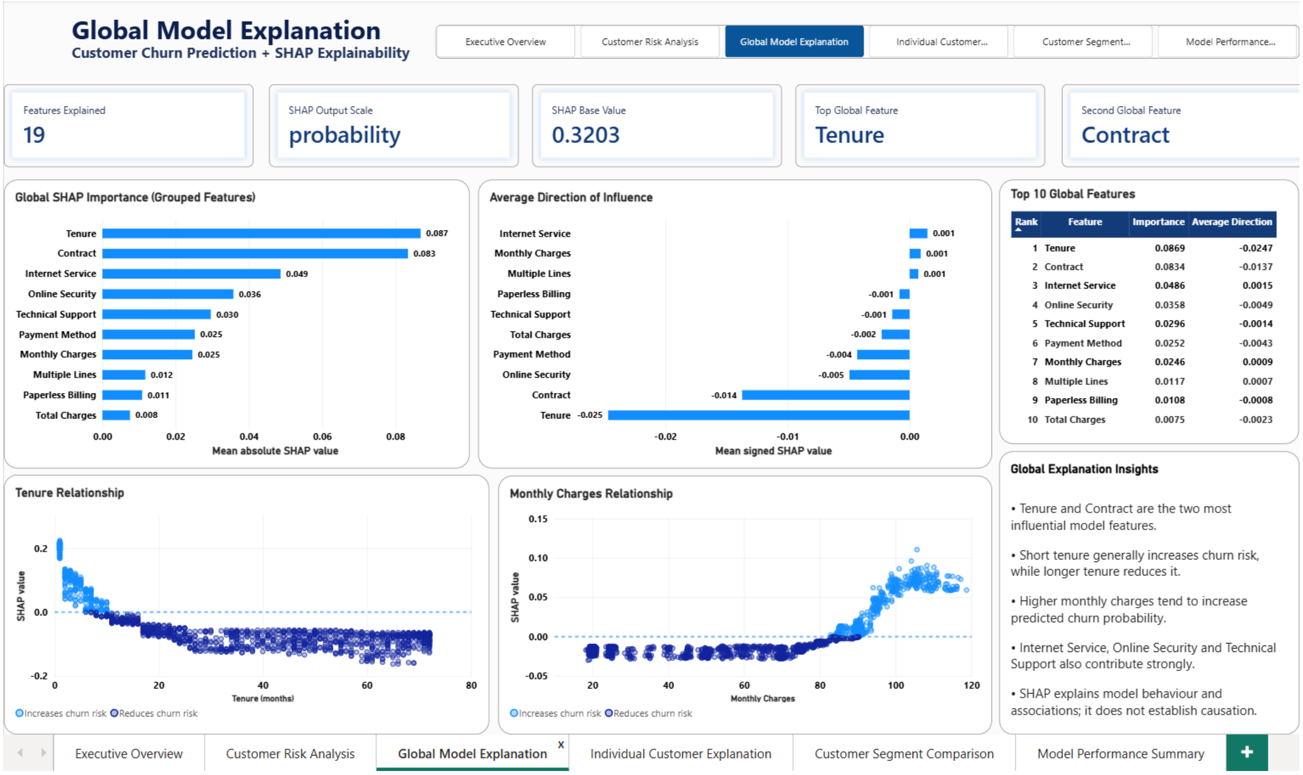

Figure 3. Static Global Model Explanation used to identify the most influential features.

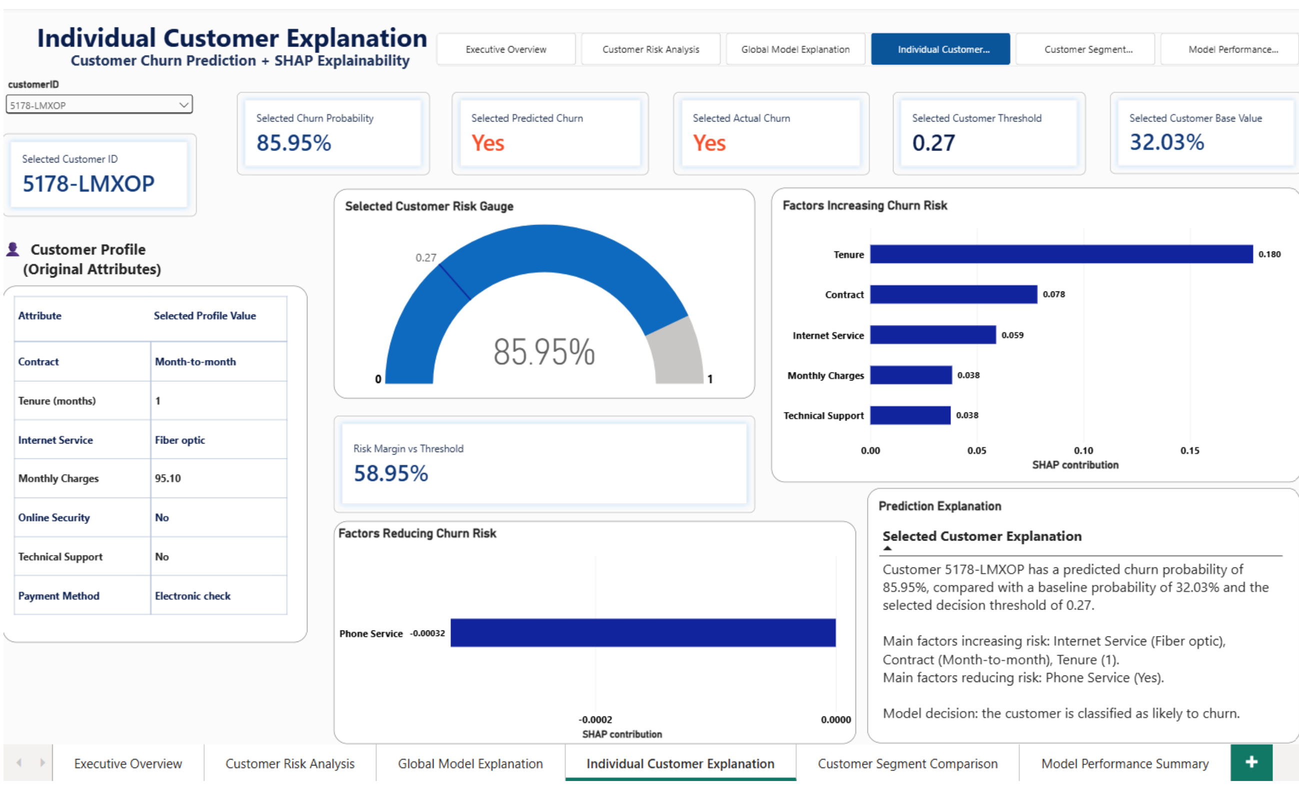

Figure 4. Static customer-level explanation showing probability, profile, and SHAP factors.

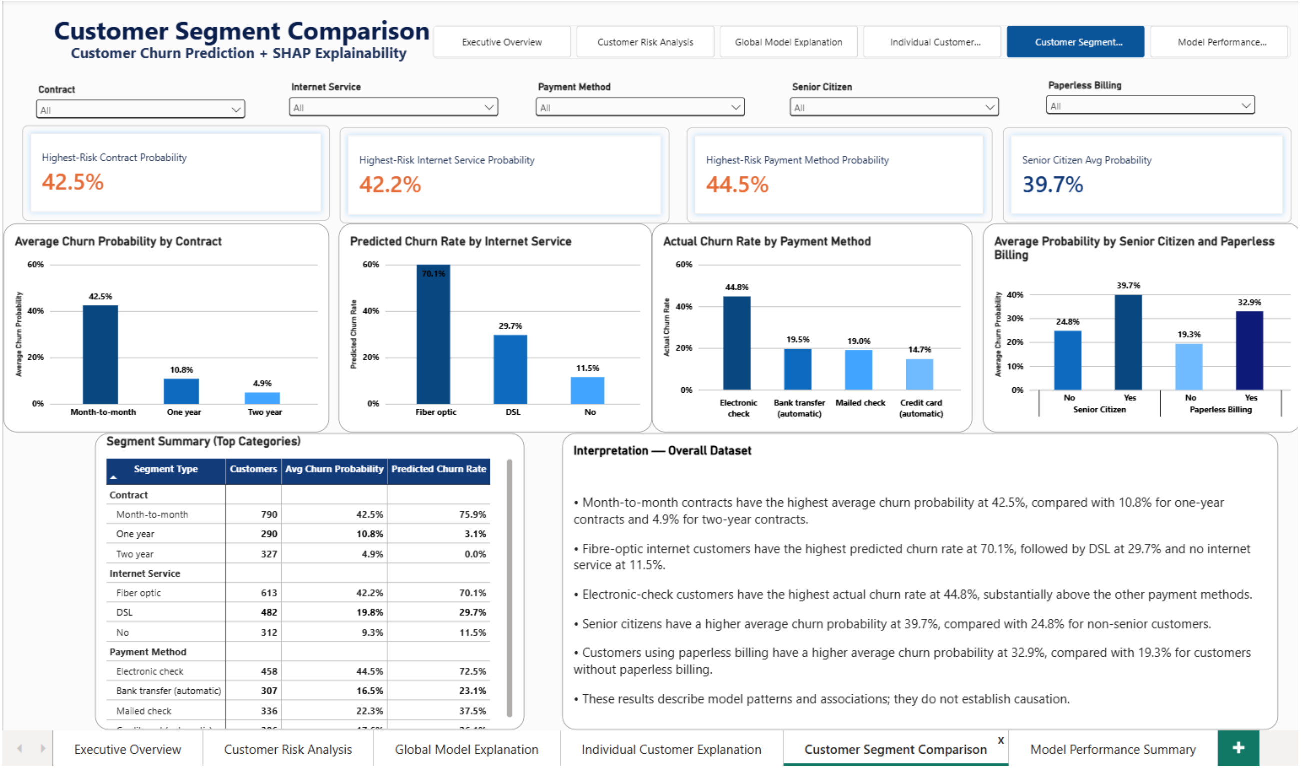

Figure 5. Static segment comparison used for contract, service, payment, and demographic questions.

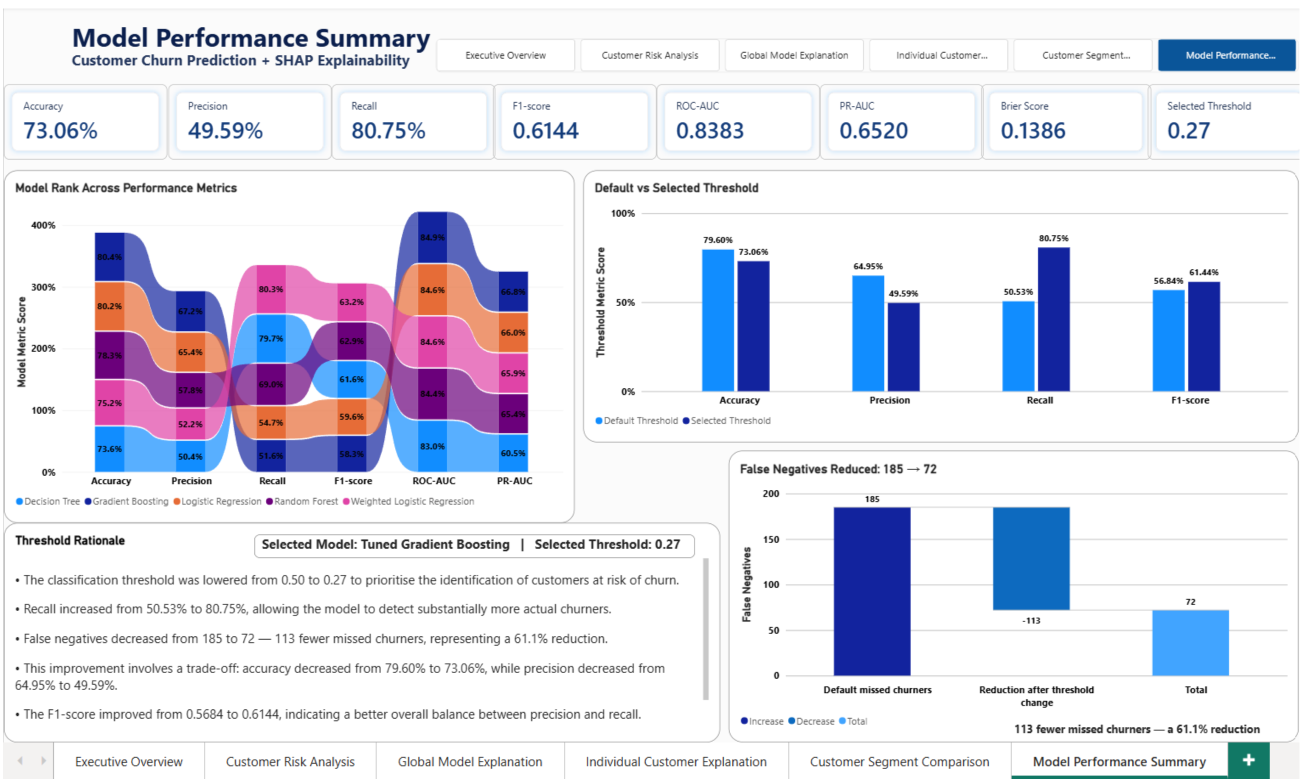

Figure 6. Static performance page used for threshold and false-negative questions.

5.1 Task structure

Six task groups were used for each condition. Each group contains two short questions, producing a maximum score of 12 points per system.

Task	Area	Purpose
1	Executive overview	Identify the number of actual and predicted churn customers.
2	Customer risk	Identify the highest-risk customer and its probability.
3	Global explanation	Identify the most important feature and its general direction of influence.
4	Individual explanation	Read the selected customer probability and strongest positive factor.
5	Segment comparison	Identify the highest-risk contract and internet-service categories.
6	Model performance	Identify the selected threshold and the reduction in false negatives.

The same subject areas were used in both conditions. In the interactive condition, participants could navigate to the relevant page and use available controls. In the static condition, answers had to be obtained from the fixed pages.

6. Data Collection Instrument

A single combined form records participant details, system order, task answers, task score, completion time, questionnaire ratings, preference, and a short comment. The same five-point agreement scale is used for both systems.

Score	Meaning
1	Strongly disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly agree

The questionnaire covers six practical dimensions: understanding, identification of churn factors, decision support, trust, ease of use, and usefulness. Task accuracy is calculated as the number of correct task points divided by 12 and expressed as a percentage.

The blank form and the participant response sheets are retained as separate companion files so that the main Phase 7 report remains concise.

7. Analysis Approach

The evaluation uses descriptive analysis because the sample is small and the main purpose is an initial system comparison. The following statistics are calculated separately for the static and interactive conditions:

- Mean task score and task accuracy
- Mean task-completion time
- Mean rating for each questionnaire dimension
- Mean overall questionnaire rating
- Number of participants preferring each system
- Recurring themes in short comments

Participant-level results are also checked to identify whether Power BI experience appears to influence completion time or ease-of-use ratings. The study does not treat the small sample as sufficient for broad population-level generalisation.

8. Quantitative Results

Verification note
The figures in this section were calculated from the completed participant response dataset.

8.1 Summary

Measure	Static baseline	Interactive dashboard
Participants	10	10
Mean task score	8.5 / 12	11.0 / 12
Mean task accuracy	70.83%	91.67%
Mean completion time	6:56	6:48
Mean overall rating	3.53 / 5	4.45 / 5

The interactive dashboard improved average task accuracy by 20.83 percentage points. The overall questionnaire rating increased by 0.92 points on the five-point scale.

8.2 Task accuracy

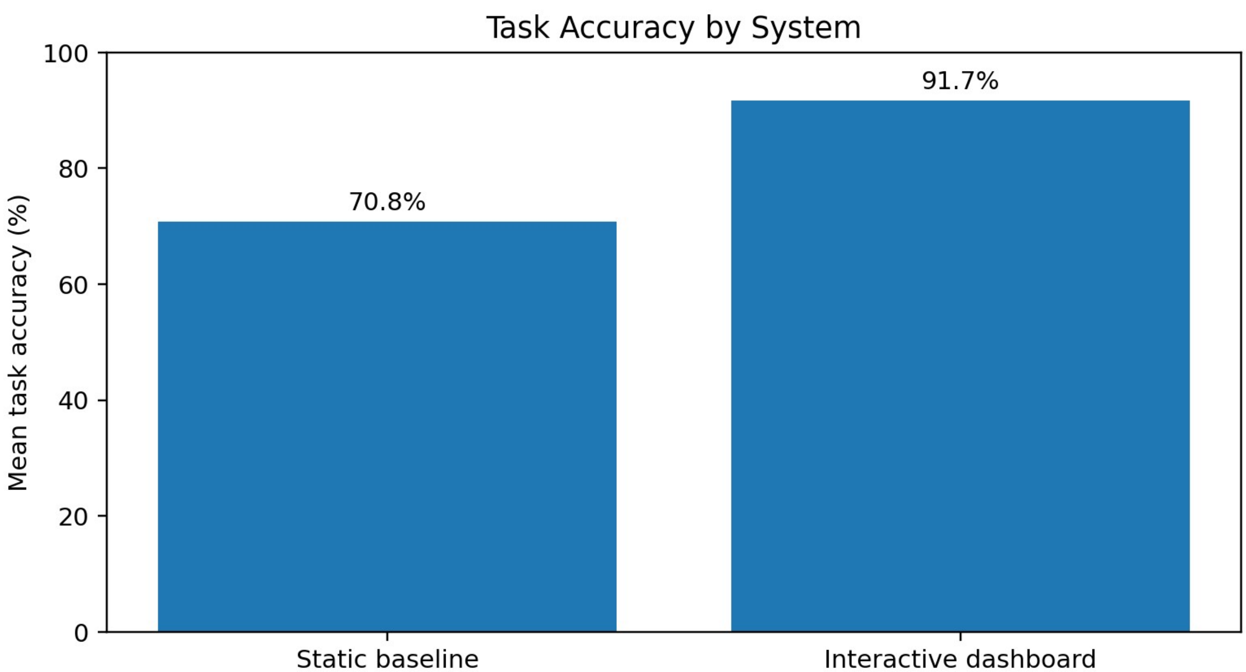

Figure 7. Average task accuracy for the two conditions.

The static baseline produced a mean accuracy of 70.83%, while the interactive dashboard reached 91.67%. The interactive condition therefore supported more correct answers across the six task areas.

8.3 Completion time

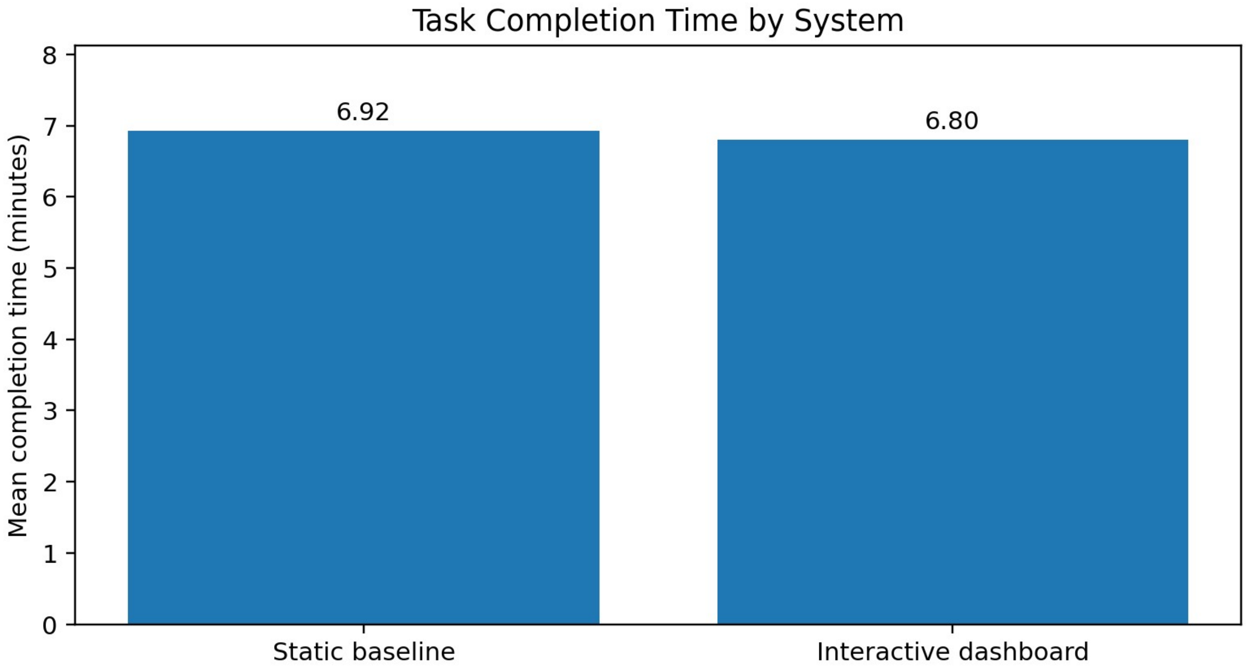

Figure 8. Mean task-completion time for the two conditions.

The interactive dashboard was 7.5 seconds faster on average. The result suggests that the available controls did not create an overall time penalty, although some users without Power BI experience required additional exploration.

8.4 Questionnaire ratings

Dimension	Static	Interactive	Difference
Understanding	3.50	4.50	+1.00
Factor identification	3.50	4.60	+1.10
Decision support	3.40	4.50	+1.10
Trust	3.40	4.20	+0.80
Ease of use	3.60	4.20	+0.60
Usefulness	3.80	4.70	+0.90

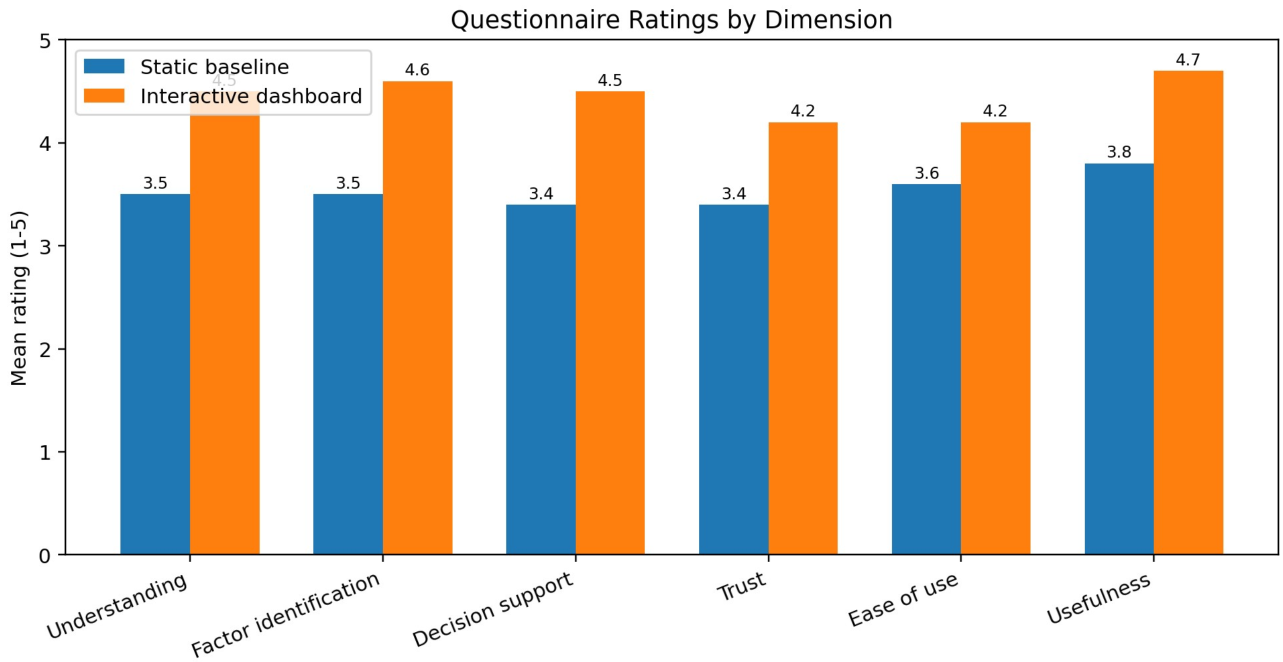

Figure 9. Mean questionnaire ratings by dimension.

The interactive dashboard received a higher average rating in every dimension. The largest gains occurred for factor identification and decision support. These dimensions depend directly on the ability to select customers, apply filters, and compare multiple views.

8.5 System preference

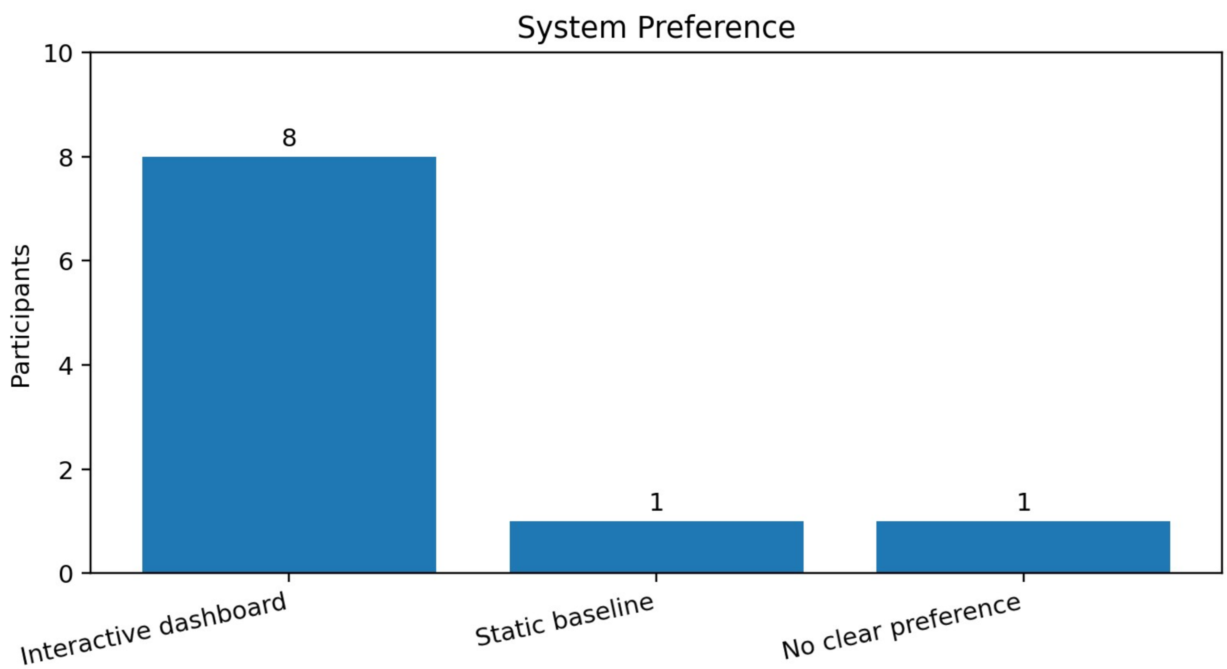

Figure 10. Final system preference.

8 participants preferred the interactive dashboard, 1 preferred the static baseline, and 1 reported no clear preference. The strongest reasons for choosing the dashboard were richer context, easier verification, and improved comparison. The static presentation was preferred by one user because it required no navigation.

9. Qualitative Findings

The short comments were grouped into recurring themes. The comments are not treated as a separate formal qualitative study; they are used to explain the numerical comparison.

Theme	Summary
Static baseline - strengths	Simple presentation, immediate access to visible values, and no interface controls to learn.
Static baseline - limitations	No customer switching, no filtering, manual comparison, and limited support for checking alternative cases.
Interactive dashboard - strengths	Dynamic customer selection, easier validation of explanations, quicker segment comparison, and stronger support for retention decisions.
Interactive dashboard - limitations	Several pages and controls required short familiarisation, especially for participants without previous Power BI experience.

Overall, the comments describe the static version as suitable for a quick review and the interactive report as more useful for detailed investigation. This distinction is consistent with the quantitative results.

10. Hypothesis Assessment

Hypothesis	Expected outcome	Assessment
H1	Higher interactive task accuracy	Supported
H2	Higher understanding	Supported
H3	Higher trust and confidence	Supported
H4	Higher ease of use and usefulness	Supported
H5	Stronger decision support	Supported
H6	Possible additional exploration time	Not supported by the overall mean

H6 was not supported by the aggregate completion time because the interactive dashboard was slightly faster in the current dataset. Participant-level results nevertheless show that users with no Power BI experience sometimes needed more time to learn the navigation.

11. Discussion

The results support the central design argument of the thesis: model explanations become more useful when users can connect high-level performance, global feature behaviour, individual customer contributions, and customer segments within one interface. The improvement was not limited to subjective preference. It was also reflected in higher task accuracy.

The dashboard was particularly helpful for questions that required movement between levels of analysis. A user could begin with the global feature ranking, select a customer, inspect the local SHAP factors, and then compare that customer with a broader segment. The static version displayed the same content but did not allow this analytical sequence to be adapted to the user’s question.

The results also indicate that interactivity should not be treated as automatically beneficial. Users need clear page titles, consistent navigation, readable labels, and a manageable number of controls. The completed report addresses this by using six purpose-specific pages rather than placing every visual on one screen.

12. Limitations

- The participant group is small, so the findings should be interpreted as an initial evaluation rather than a broadly generalisable result.
- Participants have different educational backgrounds and levels of Power BI experience, which can influence speed and perceived usability.
- The static baseline is derived from the Power BI report. Although the information is comparable, the visual style remains similar across conditions.
- The task set is intentionally short and focuses on the main functions of the dashboard rather than every possible analytical action.
- The current numerical results must be verified against completed participant forms before they are used in the submitted thesis.

13. Conclusion

The current Phase 7 comparison indicates that the interactive Power BI dashboard provides a practical advantage over a static presentation of the same model outputs. It achieved higher task accuracy, stronger ratings across all six questionnaire dimensions, and a clear preference among the participant profiles.

The largest benefits were found in the identification of churn factors, comparison of customers and segments, and support for retention decisions. The static baseline remained useful for rapid inspection, but it offered less flexibility for deeper analysis.

These findings provide initial evidence that interactive visual analytics can improve the way users interpret and apply explainable machine learning results. Once the participant sheets are confirmed, the final values can be transferred directly into the thesis results and discussion chapters.

14. Phase Deliverables and Repository Update

File	Purpose
Phase_7_1_Evaluation_Framework.pdf	Evaluation objective, research questions, hypotheses, design, participants, and measures.
Phase_7_2_Static_Explanation_Baseline_and_Evaluation_Tasks.pdf	Static dashboard pages, equivalent task sets, scoring, and answer key.
Phase_7_3_Combined_User_Evaluation_Form.pdf	Blank task, timing, questionnaire, and comparison form.
Phase_7_4_Participant_Response_Sheets.pdf	One response sheet for each participant.
Phase_7_5_Response_Analysis_and_Findings.pdf	Descriptive analysis, charts, findings, limitations, and conclusion.
Phase_7_User_Evaluation_and_Results.pdf	Consolidated Phase 7 documentation.

Recommended repository location:

- Documentation/Phase_7_User_Evaluation_and_Results.pdf
- evaluation/Phase_7_2_Static_Explanation_Baseline_and_Evaluation_Tasks.pdf
- evaluation/Phase_7_3_Combined_User_Evaluation_Form.pdf
- evaluation/Phase_7_4_Participant_Response_Sheets.pdf
- evaluation/Phase_7_Response_Data.csv
- evaluation/Phase_7_5_Response_Analysis_and_Findings.pdf

Repository status
Phase 7 is complete. The final evaluation document, participant response sheets, and analysis materials can now be added to the repository, and the README status can be changed to Completed.

Appendix A. Participant-Level Summary

ID	Name	Static accuracy	Interactive accuracy	Static time	Interactive time	Preference
P01	Lenin	83.33%	100.00%	06:10	06:35	Interactive dashboard
P02	Aahan	58.33%	83.33%	07:50	08:20	Static baseline
P03	Steve	66.67%	91.67%	07:20	07:45	Interactive dashboard
P04	Smith	58.33%	83.33%	07:30	07:55	Interactive dashboard
P05	Nathen	75.00%	100.00%	06:25	06:10	Interactive dashboard
P06	Deva	83.33%	100.00%	05:55	04:50	Interactive dashboard
P07	Deekshi	50.00%	75.00%	08:40	09:05	No clear preference
P08	Helaine	58.33%	83.33%	08:10	08:35	Interactive dashboard
P09	Akshata	91.67%	100.00%	05:40	04:25	Interactive dashboard
P10	Kshitija	83.33%	100.00%	05:35	04:20	Interactive dashboard

Appendix B. Task Answer Key

Item	Question area	Correct answer
1A	Actual churn customers	374
1B	Predicted churn customers	609
2A	Highest-risk customer	5178-LMXOP
2B	Highest churn probability	85.95%
3A	Most important global feature	Tenure
3B	General effect of longer tenure	Reduces churn risk
4A	Selected customer churn probability	85.95%
4B	Strongest positive SHAP factor	Tenure
5A	Highest-risk contract category	Month-to-month
5B	Highest predicted churn-rate internet service	Fibre optic
6A	Selected classification threshold	0.27
6B	False-negative reduction	113 fewer false negatives